

3(a)(i)	continuous;	1
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Question	Answer	Marks
3(a)(ii)	any two from: 1 smoking ; 2 air pollution ; 3 workplace pollution ; 4 previous lung disease ;	2
3(b)(i)	1 <i>ref. to</i> transcription factors ; any three from: 2 TF / gene product / protein, bind to DNA ; 3 (TF binds to) promoter ; 4 <i>ref. to</i> binding of RNA polymerase ; R if operator mentioned 5 (so) mRNA is made / transcription occurs ; ora 6 detail from Fig. 3.2 ; - gene A, switches on / controls, 4 genes or genes B and C and 2 other genes - gene B, switches on / controls, 6 genes - gene C, switches on / controls, 11 genes	4

Question	Answer	Marks
3(b)(ii)	<i>any five from:</i> 1 obtain mRNA from (basal) cell (expressing gene C) ; 2 reverse transcription of mRNA to produce cDNA ; 3 add fluorescent label to (c)DNA ; I colour 4 (microarray has) ssDNA <u>probes</u> ; 5 each from a, different / known, gene ; 6 (c)DNA hybridises / AW, to, probes / ssDNA (on microarray) ; 7 fluorescence shows the expressed genes ; 8 AVP ; e.g. <i>ref. to</i> washing off excess cDNA (after hybridisation) UV light / laser scanning (to record fluorescence pattern) intensity of fluorescence gives quantitative measure	5
3(c)	<i>any three from:</i> 1 sequence gene(s) ; 2 predict, primary structure / amino acid sequence ; 3 model receptor protein (3D) structure ; 4 find molecules, that can block receptors ; 5 without triggering effects ; 6 <i>ref. to</i> limiting receptor protein production ;	3